Grant P. Holzemer

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Experience

Google DeepMind | Machine Learning & Experimentation Platform | *March 2026 - Present*
Software Engineer III

- Proposed and led a re-architecture of XManager, the platform researchers use to launch and track ML experiments, splitting 3 critical microservices into compute-isolated zones aligned with researcher workflows so a failure in one workload no longer disrupts other researchers' experiments. The first 2 services eliminated experiment-launch load shedding and cut P99 latency by ~20%.
- Identified that the experimentation, compute, and ML serving platforms had no shared definition of a "hero run" (a critical, long-running training job); proposed and now lead a cross-platform project sending a criticality signal from XManager to compute and serving, so researchers' most important runs get prioritized compute and more precise alerting. Implementation in progress.
- Designed and built a self-serve Python observability and alerting framework for high-priority training runs (quota exhaustion, downstream failures, client errors), plus an AI bot that triages incidents; adopted by 15 research teams across Google DeepMind.
- Defined SLOs and escalation policies for the experiment analytics platform, which researchers use to analyze and compare their experiments. Drove its production readiness review, enabling handoff to SRE on-call.

Amazon Web Services | EC2 VPC | *June 2023 - March 2026*
Software Development Engineer II | *September 2025 - March 2026*
Software Development Engineer I | *January 2024 - August 2025*
Software Development Engineer Intern | *June 2023 - August 2023*

- Led design and implementation of Security Group referencing across VPC associations, letting customers reference shared security groups in rules across their VPCs; spanned multiple mutating and read-only EC2 control-plane APIs and resolved authorization, dependency-ordering, and backwards-compatibility edge cases.
- Fixed a years-old failure mode where CloudFormation Security Group updates, even description-only edits, recreated rules and briefly dropped customer production traffic; introduced batched rule updates with resilient fallbacks that eliminated these outages.
- Prevented breaking changes for 500+ customer accounts by identifying affected usage ahead of new VPC feature launches and gating rollout with targeted controls.
- Reduced large-scale consistency validation runtime by ~90% by re-implementing data access patterns with time-based indexing in DynamoDB, cutting regional data processing from months to weeks.
- Cut recovery time for a full regional outage from days to ~1 hour by automating large-scale EC2 Security Group state reprocessing.
- Built automated multi-region build and deployment pipelines that eliminated manual operator steps.
- Served as primary on-call for EC2 Security Groups and other control-plane services, resolving 400+ tickets including high-severity incidents; authored and led post-incident reviews that drove anomaly detection, rollback tooling, and better observability.
- Mentored two summer interns and multiple new hires while authoring onboarding documentation, architectural overviews, and operational runbooks to improve ramp-up time and long-term team health.

Raytheon Intelligence & Space | **Software Engineer Intern** | May 2022 - August 2022

- Built a cross-platform Electron app (Python, Angular, MySQL, AWS) tracking ~3,000 active and decayed commercial satellites in real time, with server-side STK processing to keep satellite models current.

Skills

Languages: Java, Python, C++, TypeScript, Ruby, SQL

AWS: EC2, VPC, DynamoDB, Lambda, S3, CloudFormation, CloudWatch

Google Infrastructure: Spanner, Pub/Sub, Stubby (RPC), Borg, XManager

Systems: Distributed Systems, Networking (gRPC, HTTP, TCP/UDP/ICMP, RDMA), API Design, Reliability Engineering & SLOs

Tools: Git, CI/CD Pipelines, Infrastructure as Code, Grafana, Monitoring & Alerting

Education

West Virginia University - B.S. Computer Science, Magna Cum Laude (GPA 3.78/4.0) | May 2023

Military Service and Awards

- U.S. Army National Guard, Specialist (E4), Power-Generation Equipment Repairer | June 2020 - June 2026
- Certificate of Recognition from West Virginia Governor Jim Justice
- WVU Foundation Outstanding Senior (2023), one of 52 graduating seniors university-wide recognized for scholarship, leadership, and service